

Patient Triage System

Rule-Based Clinical Triage Simulation · Educational Project

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0 1 Overview & Medical Background

What this project does

When a patient arrives in an emergency department, the first question isn't 'what's wrong?' — it's 'how urgent is this?' That sorting process is called triage, and this project simulates it in software.

A user enters patient data — symptoms, vitals, pain level — and a weighted scoring engine assigns a priority category with a list of reasons. The rules are grounded in frameworks used in real German hospitals.

Medical foundation

Manchester Triage System (MTS) — used in German and UK emergency departments since 1994. Assigns colour-coded priority levels with maximum wait times:

■ Immediate	0 min
■ Urgent	10 min
■ Normal	30 min
■ Non-Urgent	120 min

NEWS2 Score — Royal College of Physicians framework that scores physiological parameters (pulse, temperature, SpO2) to detect deterioration. This project uses NEWS2 pulse and temperature thresholds.

0 2 Inputs & Scoring Rules

Every patient goes through seven inputs. Each adds points based on severity — the total score decides the category.

Input	Trigger values	Points	Clinical reason
Critical symptoms	Chest pain, stroke, seizure...	+100 per match	Any one of these is life-threatening
Urgent symptoms	Confusion, fracture, overdose...	+30 per match	Serious but not immediately fatal
Moderate symptoms	Vomiting, UTI, ear pain...	+12 per match	Significant — patient is stable
Pulse (bpm)	>100 / >130 / >150 or <40	+20 / +40 / +60	Abnormal heart rate signals distress
Temperature (°C)	≥38 / ≥38.5 / ≥39.5 / >41	+10/+20/+35/+60	Mild fever up to dangerously high
Pain level (0–10)	≥5 moderate / ≥7 severe / ≥9	+12 / +25 / +40	Higher pain = more urgent
Shortness of breath	Yes / No	+40 if yes	Key respiratory warning sign
Age (years)	≥65 / ≥75 / ≤2	+8 / +15 / +20	Elderly and infants are higher risk

Table 1 — Scoring rules: IMMEDIATE ≥80 pts · URGENT ≥40 · NORMAL ≥15 · NON-URGENT <15

03 Implementation

File	What it does
patient.py	Patient class — holds all data for one person as a Python object
triage_rules.py	Scoring engine — assign_triage() runs all rules and returns the category
test_data.py	Four pre-built sample patients to verify the system works correctly
main.py	Terminal interface — manual patient entry or automated test run
index.html	Standalone web page — full triage UI runs in any browser, no server needed

Table 2 — Project file structure

Patient data model

Each patient is a Python object — keeping data separate from logic is standard practice in medical software.

```
class Patient:
    def __init__(self, name, age,
                  symptoms, pain_level,
                  fever, pulse, sob):
        self.symptoms = [s.lower()
                          for s in symptoms]
        self.triage_category = None
        self.triage_reason = []
```

Scoring engine

assign_triage() works through symptoms, vitals, and age — adding points at each step. One comparison at the end decides:

```
if score >= 80: category = 'IMMEDIATE'
elif score >= 40: category = 'URGENT'
elif score >= 15: category = 'NORMAL'
else: category = 'NON-URGENT'
```

Web interface (index.html)

A single file — open it in Chrome and everything works with no installation. All triage logic is rewritten in JavaScript inside the same file.

- Symptoms checklist — 4 tabs (Critical / Urgent / Moderate / Minor)
- Selected symptoms shown as pills before submitting
- Colour-coded result card matching triage category
- Session log table of all patients triaged so far
- One-click CSV export — opens directly in Excel

Terminal mode

The Python CLI offers two modes — enter a patient step by step, or run all four test patients automatically:

```
python main.py

1 – Enter a patient manually
2 – Run sample test patients
```

04 Test Results

Four patients were tested — one per category. Results were verified by manually tracing each score.

Ahmed	67	Chest pain, dizziness	37.2°C	140	9/10	Yes	220+	IMMEDIATE
Klaus	82	Confusion, weakness	39.1°C	105	3/10	No	~78	URGENT
Sara	34	Headache, nausea	38.0°C	88	4/10	No	~22	NORMAL
Lena	22	Sore throat, cough	37.5°C	72	2/10	No	~4	NON-URGENT

Table 3 — Ahmed: chest pain +100, pulse 140 +40, pain 9/10 +40, SOB +40 → total 220 → IMMEDIATE ✓

05 Discussion

What worked well

The scoring logic is fully transparent — every point has a reason and every decision can be traced back to the inputs. No black boxes.

Rules are grounded in MTS and NEWS2, not made up.

Dual interface (CLI + web) makes it easy to test and present.

CSV export bridges the gap between a triage session and usable data.

Honest limitations

- No patient history, allergies, or comorbidities.
- Shortness of breath is binary — real triage grades severity.
- SpO2 not captured — a key NEWS2 parameter.
- Fixed symptom list — unusual cases can't be entered freely.
- No HL7 FHIR integration for real health IT interoperability.

vs. a real triage nurse: A trained MTS nurse follows structured discriminator flowcharts, observes body language, asks follow-up questions, and draws on years of experience. A scoring formula can't replicate that — but it can teach the principles behind it.

06 Future Improvements

Next step	What it would add
Add SpO2 as a direct input	Completes the full NEWS2 vital sign set — the most important missing piece
SQLite database	Persistent patient records across sessions — more realistic for demos
Flask / FastAPI backend	Multi-user access and proper server-side logging
HL7 FHIR integration	Speaks the language of real hospital information systems
ML-based triage classifier	Replace fixed rules with a model learned from real triage data
German / Arabic language support	More realistic for diverse clinical environments in Bavaria and beyond

Table 4 — Proposed next steps

07 Conclusion

Building this system made the triage process feel real in a way that textbooks don't. Writing the rules — deciding that a pulse over 150 adds 60 points, or that an elderly patient automatically scores higher — required genuine engagement with the clinical reasoning behind each decision.

The result is a working simulation: a Python back-end with clean object-oriented design, a browser-based interface that needs no installation, and a CSV export ready for analysis. It's not a medical device — but it demonstrates exactly the kind of thinking that goes into building one. Health informatics is about understanding both the clinical and the technical well enough to make them work together. This was a first real attempt at that.

■ ■ This is an educational simulation only. It does not constitute medical advice and must never be used to make real clinical decisions. All triage assessments must be performed by qualified medical professionals.

08 References

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